

# PRANJALYA TIWARI

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## PROFILE

Results-driven Senior AI / Machine Learning Engineer with 5+ years of experience engineering high-throughput Agentic AI systems, custom LLM toolchains, deep learning model development and distributed MLOps pipelines. Proven track record of bridging low-level system performance (Rust, PyTorch) with modern agentic architectures (MCP, LangGraph) to drive product innovation and operational efficiency.

## EDUCATION

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| <b>Bhilai Institute of Technology</b><br>B.Tech. Computer Science and Engineering ( <i>GPA: 9.27/10</i> ) | Durg, CG<br>August 2017 - June 2021 |
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## EXPERIENCE

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| <b>Ksolves India Ltd.</b><br><i>Senior Machine Learning Engineer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Noida, UP<br>January 2025 - Present    |
| <ul style="list-style-type: none"><li>Architected multi-agent LangGraph architectures for enterprise clients (<b>Nokia</b>), enabling autonomous insight generation from unstructured internal and external data sources.</li><li>Custom-extended <b>Langflow</b> with Model Context Protocol (MCP) integrations, custom SSO, and granular RBAC to empower non-technical teams to securely prototype enterprise agents.</li><li>Engineered an enterprise Text-to-Code data analysis pipeline using LangGraph and Chainlit, automating PowerBI report extraction and execution, reducing report turnaround time by 80% for enterprise users.</li><li>Fine-tuned Small Language Models (SLMs) and aligned agent intent refinement modules to maximize domain-specific analytical accuracy.</li></ul> |                                        |
| <b>Deepsync Technologies / Dubpro.ai</b><br><i>Machine Learning Engineer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Noida, UP<br>July 2021 - December 2024 |
| <ul style="list-style-type: none"><li>Built and deployed enterprise Conversational AI, speech, and vision models in production environments.</li><li>Designed an end-to-end multi-modal dubbing pipeline on Apache Airflow, processing hundreds of concurrent video streams with fault-tolerant execution.</li><li>Optimized inference pipelines and swapped costly proprietary cloud APIs with optimized open-source alternatives, reducing infrastructure OpEx by 30–40% with models on Huggingface and custom infra.</li><li>Developed interactive Gradio/Streamlit prototypes to showcase technical capabilities directly to senior leadership and enterprise prospects.</li></ul>                                                                                                             |                                        |

## SKILLS

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|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI &amp; Systems Architecture:</b> | Agentic AI, Model Context Protocol (MCP), RAG, Intent Refinement, Vector Search, MLOps, Reinforcement Learning, Computer Vision, Speech AI |
| <b>Languages &amp; Frameworks:</b>    | Python, Rust, C++, Golang, PyTorch, LangChain, LangGraph, LlamaIndex, WebAssembly (WASM), TensorFlow, Scikit-learn                         |
| <b>Infrastructure &amp; Cloud:</b>    | NVIDIA Triton Server, Docker, CUDA, Qdrant Vector DB, Apache Airflow, AWS, GCP, Azure, Linux/Shell, CI/CD, FastAPI, Node.js, Next.js       |
| <b>Certifications &amp; Honors:</b>   | NeurIPS Published Author, NPTEL C++ (Gold Medal), Deep Learning Specialization (Coursera), RL Specialization (U. Alberta)                  |

## FEATURED PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |
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| <b>AstraChaos 3D   Interactive Celestial Mechanics &amp; Agentic Copilot</b><br><i>Tech Stack: Rust, WebAssembly, Python FastAPI, NVIDIA Nemotron, MCP, Three.js, Recharts</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | [Demo] [Code] |
| <ul style="list-style-type: none"><li><b>Generative Inverse Physics Engine:</b> Built a vectorized Differential Evolution optimizer over an 18D state space (<math>\mathbb{R}^{18}</math>) to reverse-engineer chaotic initial condition topologies from natural language, streaming real-time SSE updates.</li><li><b>Agentic Microservice &amp; MCP Tooling:</b> Engineered a Python FastAPI service connecting NVIDIA Nemotron with Model Context Protocol (MCP) tools for intent parsing and automated state vector synthesis with zero-downtime fallbacks.</li><li><b>Rust/WASM Core &amp; Visual Analytics:</b> Developed an RK4 numerical integrator in Rust/WASM piping zero-allocation memory directly to Three.js at 120 FPS to render parallel perturbed timelines (<math>\delta = 10^{-7}</math>) with Hamiltonian energy scorecards.</li></ul> |               |

## Ghazal / Poem Semantic Search & RAG System

[Demo]

*Tech Stack: Python, Llama 3.2, Qdrant Vector DB, LangChain, FastEmbed*

- Built an end-to-end semantic search and retrieval-augmented generation engine using Qdrant and fine-tuned embeddings for precise domain-specific intent matching and low-latency response generation.

## Multi-Lingual NLP Service Deployment on NVIDIA Triton

*Tech Stack: Python, Docker, NVIDIA Triton Inference Server, Azure MLOps*

- Containerized and deployed high-throughput multi-lingual NLP models (NER, Summarization, Classification) on Triton Server, optimizing GPU memory allocation and batching efficiency.

## Multi-Stage LLM Pipeline for Conversational AI

*Tech Stack: Python, PyTorch, LlamaIndex, Transformers*

- Designed a multi-stage generation framework incorporating real-time emotion classification and multi-turn context awareness to refine conversational intent and similarity scoring.

## Flow Forecast (Open-Source Contributor)

[Code]

*Tech Stack: PyTorch, Time-Series Modeling, Distributed Training*

- Contributed SOTA deep learning models and interpretability metrics to an open-source deep learning repository for multi-dimensional time series forecasting.

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## PUBLICATIONS & AWARDS

### NeurIPS Workshop Publication

December 2020

Co-authored "*FlowDB: A new large scale river flow and flash flood dataset*", contributing novel multi-dimensional hydrological datasets for quantitative climate modeling.

### Smart India Hackathon Finalist (Team Leader)

July 2020

Led a team of engineers to build deep learning toxicity classification models on social media text stream data.

### Winner - BITSHINE Hardware & Software Innovation Expo

November 2018

Secured 1st place for architecting a scalable prototype web platform tailored for digital health resources.